02/11/2026 : $\boxed{\text{lec 9}}$

QR iteration: A is $n \times n$. $A_0 = A$ at k^{th} step,

$Q_k^T = Q_k^{-1}$

- compute $A_k = Q_k R_k$ (Q_k orthogonal, R_k upper Δ)
- let $A_{k+1} = R_k Q_k$

$\{A_k\}$ similar : same eigenvalues

under certain conditions $A_k \to \bar{A}$ (Schur form??)

Gram Schmidt process! $\leftarrow$ has some cons

Step 1: use Householder Reflection to get an upper Hessenberg matrix

$$P_{n-2} \cdots P_2 P_1 A P_1 P_2 \cdots P_{n-2} = H^{(0)}$$

Step 2: Apply rotation to $H^{(0)}$ $\quad H^{(k)} \overset{?}{=} R_k \cdots R_3 R_2 R_1 H^{(0)}$??

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \text{ in } 2D$$

$$H = \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{pmatrix}$$

rot. $\vec{e}_1$ $\quad$ rot. $\vec{e}_2$ $\quad$ rot. $\vec{e}_3$

$\vec{v} = \begin{pmatrix} * \\ * \end{pmatrix} \in \mathbb{R}^2$

$\|\vec{v}\| \vec{e}_1 = \begin{pmatrix} * \\ 0 \end{pmatrix}$

Again start with Hessenberg matrix

$$\underbrace{\left(\begin{array}{c|c} R_{-\theta_1} & 0 \\ \hline 0 & I_2 \end{array}\right)}_{R_1} \left(\begin{array}{cccc} \overset{\vec{v}}{\boxed{\begin{matrix}**\end{matrix}}} & * & * & * \\ & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right) = \left(\begin{array}{cccc} * & \vec{w} & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right)$$

Apply rotation $\vec{w}$ $\quad R_{\theta_2}\vec{w} = \|\vec{w}\|\,\vec{e}_2$

$$\underbrace{\left(\begin{array}{c|c|c} 1 & & \\ \hline & R_{-\theta_2} & \\ \hline & & 1 \end{array}\right)}_{R_2} \left(\begin{array}{cccc} * & \vec{w} & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{array}\right)$$

apply $R_{-\theta_3}$ s.t. $R_{-\theta_3}\vec{u} = \|\vec{u}\|\,\vec{e}_3$

$$\left(\begin{array}{c|c} I_2 & \\ \hline & R_{-\theta_3} \end{array}\right) \left(\begin{array}{cccc} * & \vec{w} & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{array}\right) = \left(\begin{array}{cccc} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{array}\right)$$

$R_3 R_2 R_1 H = R$ upper triang.

$$\Rightarrow H^{(0)} = \underbrace{\left[R_1^{-1} R_2^{-1} R_3^{-1}\right]}_{Q^{(0)}} R^{(0)} \quad ; \quad H^{(1)} = Q^{(0)} R^{(0)}$$

$Q^{(0)}$ orthogonal!

$$H^{(1)} = R^{(0)}Q^{(0)} = R_3 R_2 R_1 H^{(0)} R_1^{-1} R_2^{-1} R_3^{-1}$$

$$= \underbrace{\boxed{R^{(0)} R_1^{-1}}}_{\text{upper triang.}} \underbrace{R_2^{-1} R_3^{-1}}_{\text{Hess.}}$$

$$R^{(0)}R_1^{-1} = \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} \left(\begin{array}{c|c} \boxed{R_{\theta_1}} & \\ \hline & I \end{array}\right) = \begin{pmatrix} \boxed{\begin{matrix} * & * \\ * & * \end{matrix}} & \begin{matrix} * & * \\ * & * \end{matrix} \\ \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} & \begin{matrix} * & * \\ 0 & * \end{matrix} \end{pmatrix}$$

continuing $\Longrightarrow$ $H^{(1)}$ will be upper Hessenberg matrix.

<u>last class</u>: showed how to take $A \longrightarrow H^{(0)}$: Hessenberg.

$$A = \begin{pmatrix} 4 & 1 & -2 & 2 \\ 1 & 2 & 0 & 1 \\ -2 & 0 & 3 & -2 \\ 2 & 1 & -2 & -1 \end{pmatrix} \longrightarrow H^{(0)} = \begin{pmatrix} \boxed{\begin{matrix} 4 \\ 3 \end{matrix}} & \begin{matrix} 3 \\ 10/3 \end{matrix} & \begin{matrix} 0 \\ 5/3 \end{matrix} & \begin{matrix} 0 \\ 0 \end{matrix} \\ 0 & 5/3 & -33/25 & -68/75 \\ 0 & 0 & -68/75 & 149/75 \end{pmatrix}$$

$\cos(-\theta_1) = 4/5$, $\sin(-\theta_1) = 3/5$

$$R_1 = \begin{pmatrix} 4/5 & 3/5 & 0 & 0 \\ -3/5 & 4/5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad H^{(0)} = \begin{pmatrix} 5 & 22/5 & 1 & 0 \\ 0 & \boxed{13/15} & 4/3 & 0 \\ 0 & \boxed{5/3} & -33/25 & -68/75 \\ 0 & 0 & -68/75 & 149/75 \end{pmatrix}$$

$\cos(-\theta_2) = 13/15 \approx 0.46$, $\sin(-\theta_2) = 5/3 \approx 0.89$

$$R_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.46 & 0.89 & 0 \\ 0 & -0.89 & 0.46 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (R_1 H^{(0)}) = \begin{pmatrix} 5 & 4.4 & 1 & 0 \\ 0 & 1.8 & -0.56 & -0.8 \\ 0 & 0 & -1.79 & -0.41 \\ 0 & 0 & -0.9 & 1.99 \end{pmatrix}$$

$H^{(k)} \to H^*$ which is diagonal! so we get eigenvalues.

Iterative Methods

Solve $Ax = b$: generate a seq of approx. solutions x_k

s.t. $\lim_{k\to\infty} x_k = x$

$Ax = b \iff x = Bx + c$ equivalent linear system

$x^{(k+1)} = \underbrace{B}_{\text{iter matrix}} x^{(k)} + c$ fixed-point iteration

"for linear system how do we find an equivalent fixed-point iteration"

<u>idea</u>: linear operator splitting $A = P - R$, where P is easily invert

Begin $A\vec{x} = \vec{b}$ suppose $A = P - R$ s.t. P is "easy to invert"

$$\Rightarrow A\vec{x} = \vec{b} \Rightarrow (P - R)\vec{x} = \vec{b}$$
$$\Rightarrow P\vec{x} = \vec{b} + R\vec{x}$$
$$\vec{x} = P^{-1}(Rx + \vec{b})$$
$$= \underbrace{P^{-1}R}_{B} x + \underbrace{P^{-1}\vec{b}}_{\vec{c}}$$

$$\Rightarrow \vec{x} = B\vec{x} + \vec{c}$$

$\boxed{\text{Jacobi Method}}$ split $A = P - R$. Assume $A = D + L + U$

we set $x^{(k+1)} = -D^{-1}(L+U)x^{(k)} + D^{-1}b$

<u>Example</u> $\begin{cases} 2x_1 - x_2 = 1 \\ -x_1 + 2x_2 = 1 \end{cases} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \vec{x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{x}^{(1)} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \vec{x}^{(0)} + \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad x^{(1)} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}; \quad x^{(2)} = \begin{pmatrix} 3/4 \\ 3/4 \end{pmatrix}, \quad x^{(3)} = \begin{pmatrix} 7/8 \\ 7/8 \end{pmatrix}$$

<u>def</u>: A is <u>strictly diagonal dominant</u> if

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

$$\|A\|_\infty = \sup_{x \in \mathbb{R}^n} \frac{\|Bx\|_\infty}{\|x\|_\infty}$$

<u>thm</u>: If A is strictly diagonal dominant, then $\|B_J\|_\infty < 1$, where $B_J = -D^{-1}(L+U)$ $\therefore$ Jacobi converges.

Gauss-Seidel Method

Assume $A = D + L + U$

iter matrix

$$x^{(k+1)} = \boxed{-(L+D)^{-1} U} x^{(k)} + (L+D)^{-1} b$$